

Information Systems

Data Modeling

IDENTIFICATION

CODE : IF-3-MD
ECTS : 2.0

HOURS

Lectures : 12.0 h
Seminars : 14.0 h
Laboratory : 0.0 h
Project : 0.0 h
Teacher-student
contact : 26.0 h
Personal work : 30.0 h
Total : 56.0 h

ASSESSMENT METHOD

Written exam (1h30)

TEACHING AIDS

Lecture notes for the course, work
sheets for classwork (in French)

TEACHING LANGUAGE

French

CONTACT

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AIMS

The goal of this course is to gain theoretical understanding of the principles of data management.

It consists of two main parts: relational query languages (algebraic, logic and SQL) for understanding the basics optimization process of SQL queries.

The second focuses on constraints, mainly functional dependencies and reasoning related to database design.

Expected skills:

- Writing equivalent queries in several languages
- Reasoning on equivalent query languages
- Understanding the different types of data model (conceptual, logical)
- Transforming schema between different data models
- Mastering database design with quality guarantees

CONTENT

- Introduction to data models.
- Applications to databases, information systems, software engineering and artificial intelligence
- Relational data model
 - * Structure.
 - * Languages: relational algebra, domain relational calculus and Datalog
 - * Constraints: functional dependencies, inclusion dependencies, keys, foreign keys
- Extended Entity-Relationship (ER) models
- Database design:
 - * Conceptual ER
 - * Universal relation
 - * Armstrong databases

BIBLIOGRAPHY

- [1] Abiteboul, Rick Hull, Victor Vianu, Foundations of databases, 685 pages, 1995, Addison-Wesley
- [2] Carlo Batini, Stefano Ceri, Shamkant B. Navathe, Conceptual Database Design: An Entity-Relationship Approach, 455 Pages, Benjamin/Cummings
- [3] Mark Levene, Georges Loizou, Guided tour of relational databases and beyond, 625 pages, 1999, Springer
- [4] Heikki Mannila, Kari-Jouko Raiha, The Design of Relational Databases, 2nd edition, 1994, Addison-Wesley

PRE-REQUISITE

Set theory, Algorithmics basic notions